Homework 2

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```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
## Warning: package 'tibble' was built under R version 4.5.2
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
## Warning: package 'purrr' was built under R version 4.5.2
```

```
## Warning: package 'lubridate' was built under R version 4.5.2
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v readr     2.2.0
## v ggplot2   4.0.3     v stringr   1.6.0
## v lubridate 1.9.5     v tibble    3.3.1
## v purrr     1.2.2     v tidyr     1.3.2
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(nlme)
```

```
## Warning: package 'nlme' was built under R version 4.5.2
```

```
##
## Attaching package: 'nlme'
##
## The following object is masked from 'package:dplyr':
##
##     collapse
```

#1.Loading tlc dataset

```r
#install.packages("haven")
library(haven)
library(nlme)

tlc <- read_sas("tlc.sas7bdat")
```

#2. Select just the placebo group (trt = P)

```r
just_placebo_long <- tlc %>%
  subset(trt == "P") %>%
  pivot_longer(starts_with("y"), names_to = "time", values_to = "y")
just_placebo_long
```

```
## # A tibble: 200 x 4
##       id trt   time      y
##    <dbl> <chr> <chr> <dbl>
##  1     1 P     y0     30.8
##  2     1 P     y1     26.9
##  3     1 P     y4     25.8
##  4     1 P     y6     23.8
##  5     4 P     y0     24.7
##  6     4 P     y1     24.5
##  7     4 P     y4     22
##  8     4 P     y6     22.5
##  9     7 P     y0     28.6
## 10     7 P     y1     20.8
## # i 190 more rows
```

#3. Fit incorrect general linear model

```r
wrong_model <- lm(y ~ relevel(as.factor(time), 4), data = just_placebo_long)
summary(wrong_model)
```

```
##
## Call:
## lm(formula = y ~ relevel(as.factor(time), 4), data = just_placebo_long)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.146  -3.849  -1.066   3.579  19.654
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    23.6460     0.7745  30.530   <2e-16 ***
## relevel(as.factor(time), 4)y0   2.6260     1.0953   2.397   0.0174 *
## relevel(as.factor(time), 4)y1   1.0140     1.0953   0.926   0.3557
## relevel(as.factor(time), 4)y4   0.4240     1.0953   0.387   0.6991
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.477 on 196 degrees of freedom
## Multiple R-squared:  0.0327, Adjusted R-squared:  0.0179
## F-statistic: 2.209 on 3 and 196 DF,  p-value: 0.08836
```

```
anova(wrong_model)
```

```
## Analysis of Variance Table
##
## Response: y
##                               Df Sum Sq Mean Sq F value  Pr(>F)
## relevel(as.factor(time), 4)    3  198.7  66.247  2.2088 0.08836 .
## Residuals                    196 5878.6  29.993
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

##R Results F statistic: 2.2088 DOF: 196 P-value: 0.088

Fail to reject the null hypothesis; there's insufficient evidence to say that the means differ across the different time points.

##SAS Results

F statistic: 2.21 DOF: 196 P-value: 0.0884

Fail to reject the null hypothesis; there's insufficient evidence to say that the means differ across the different time points.

#4. Fit generalized linear model

```
right_model <- gls(y ~ relevel(as.factor(time), 4),
                   data = just_placebo_long,
                   correlation = corSymm(form = ~ 1|id),
                   weights = varIdent(form = ~1|time),
                   method = "ML"
                   )
summary(right_model)
```

```
## Generalized least squares fit by maximum likelihood
##   Model: y ~ relevel(as.factor(time), 4)
##   Data: just_placebo_long
##        AIC      BIC    logLik
##   1063.102 1109.278 -517.5509
```

```
##
## Correlation Structure: General
##  Formula: ~1 | id
##  Parameter estimate(s):
##  Correlation:
##   1     2     3
## 2 0.829
## 3 0.839 0.861
## 4 0.756 0.759 0.870
## Variance function:
##  Structure: Different standard deviations per stratum
##  Formula: ~1 | time
##  Parameter estimates:
##       y0       y1       y4       y6
## 1.000000 1.086993 1.145098 1.122537
##
## Coefficients:
##                                Value Std.Error   t-value p-value
## (Intercept)                   23.646 0.7975874 29.646908  0.0000
## relevel(as.factor(time), 4)y0  2.626 0.5331653  4.925302  0.0000
## relevel(as.factor(time), 4)y1  1.014 0.5452260  1.859779  0.0644
## relevel(as.factor(time), 4)y4  0.424 0.4115300  1.030302  0.3041
##
##  Correlation:
##                               (Intr) r(.(),4)0 r(.(),4)1
## relevel(as.factor(time), 4)y0 -0.489
## relevel(as.factor(time), 4)y1 -0.387  0.671
## relevel(as.factor(time), 4)y4 -0.219  0.586     0.646
##
## Standardized residuals:
##        Min         Q1        Med         Q3        Max
## -1.8172661 -0.7001058 -0.2006642  0.6410404  3.5202591
##
## Residual standard error: 4.973654
## Degrees of freedom: 200 total; 196 residual
```

```
anova(right_model)
```

```
## Denom. DF: 196
##                              numDF   F-value p-value
## (Intercept)                      1 1396.6946  <.0001
## relevel(as.factor(time), 4)      3   10.2321  <.0001
```

##R Results

F statistic: 10.2321 DOF: 196 P-value: <0.0001

Reject the null hypothesis with the P value under 0.05, so there is evidence for the means differing across time points.

##SAS Results

F statistic: 10.23 DOF: 196 P-value: <0.001

#5. Question about what a piece of output from SAS and nlme in R is about.

For the nlme summary, I understood most of the output, but I wasn't entirely sure what the "Formula: ~1 | time" output meant.

From SAS, the entire "Iterations History" table was something I didn't recognize. Iteration 1 or 0 didn't seem to match up to anything in the data or the tests we were running.

#6. Revert whole dataset to include trt = A as well as P

```
tlc_longer <- tlc %>%
  pivot_longer(starts_with("y"), names_to = "time", values_to = "y")
```

#7. Fit model with both groups and follow the process listed in Canvas.

```
#same thing as question 3, just with tlc_longer
model1 <- gls(y ~ relevel(as.factor(time), 4) * relevel(as.factor(trt), 2),
              data = tlc_longer,
              correlation = corSymm(form = ~ 1 | id),
              weights = varIdent(form = ~ 1 | time),
              method = "ML"
              )

summary(model1)
```

```
## Generalized least squares fit by maximum likelihood
##   Model: y ~ relevel(as.factor(time), 4) * relevel(as.factor(trt), 2)
##   Data: tlc_longer
##        AIC      BIC    logLik
##   2461.368 2533.214 -1212.684
##
## Correlation Structure: General
##  Formula: ~1 | id
##  Parameter estimate(s):
##  Correlation:
##   1     2     3
## 2 0.571
## 3 0.570 0.775
## 4 0.577 0.582 0.581
## Variance function:
##  Structure: Different standard deviations per stratum
##  Formula: ~1 | time
##  Parameter estimates:
##       y0       y1       y4       y6
## 1.000000 1.325866 1.370446 1.524806
##
## Coefficients:
##                                                              Value Std.Error
## (Intercept)                                                 23.646 1.0830657
## relevel(as.factor(time), 4)y0                                2.626 0.8885231
## relevel(as.factor(time), 4)y1                                1.014 0.9343055
## relevel(as.factor(time), 4)y4                                0.424 0.9464420
## relevel(as.factor(trt), 2)A                                 -2.884 1.5316862
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A  3.152 1.2565614
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A -8.254 1.3213074
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A -5.672 1.3384711
```

```
##                                                      t-value p-value
## (Intercept)                                        21.832470  0.0000
## relevel(as.factor(time), 4)y0                       2.955466  0.0033
## relevel(as.factor(time), 4)y1                       1.085298  0.2785
## relevel(as.factor(time), 4)y4                       0.447994  0.6544
## relevel(as.factor(trt), 2)A                        -1.882892  0.0605
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A  2.508433  0.0125
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A -6.246843  0.0000
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A -4.237671  0.0000
##
##  Correlation:
##                                                      (Intr) r(.(),4)0
## relevel(as.factor(time), 4)y0                       -0.758
## relevel(as.factor(time), 4)y1                       -0.573  0.624
## relevel(as.factor(time), 4)y4                       -0.547  0.607
## relevel(as.factor(trt), 2)A                         -0.707  0.536
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A  0.536 -0.707
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A  0.405 -0.441
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A  0.387 -0.429
##                                                      r(.(),4)1 r(.(),4)4
## relevel(as.factor(time), 4)y0
## relevel(as.factor(time), 4)y1
## relevel(as.factor(time), 4)y4                        0.767
## relevel(as.factor(trt), 2)A                          0.405     0.387
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A -0.441    -0.429
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A -0.707    -0.542
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A -0.542    -0.707
##                                                      r(.()2 r(.(),4)02
## relevel(as.factor(time), 4)y0
## relevel(as.factor(time), 4)y1
## relevel(as.factor(time), 4)y4
## relevel(as.factor(trt), 2)A
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A -0.758
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A -0.573  0.624
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A -0.547  0.607
##                                                      r(.(),4)12
## relevel(as.factor(time), 4)y0
## relevel(as.factor(time), 4)y1
## relevel(as.factor(time), 4)y4
## relevel(as.factor(trt), 2)A
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A  0.767
##
## Standardized residuals:
##        Min         Q1        Med         Q3        Max
## -2.1977297 -0.6919512 -0.1530936  0.5347909  5.6899331
##
## Residual standard error: 4.972081
## Degrees of freedom: 400 total; 392 residual
```

```
anova(model1)
```

```
## Denom. DF: 392
```

```
##                                                     numDF   F-value p-value
## (Intercept)                                             1 2583.7140  <.0001
## relevel(as.factor(time), 4)                             3   61.4946  <.0001
## relevel(as.factor(trt), 2)                              1    4.2268  0.0405
## relevel(as.factor(time), 4):relevel(as.factor(trt), 2)  3   35.9299  <.0001
```

We're seeing significance, which means we reject the null hypothesis; there IS a difference over time. This trend over time is NOT parallel.

```
# 1. Look at the parameter estimates from model1
summary(model1)
```

```
## Generalized least squares fit by maximum likelihood
##   Model: y ~ relevel(as.factor(time), 4) * relevel(as.factor(trt), 2)
##   Data: tlc_longer
##        AIC      BIC    logLik
##   2461.368 2533.214 -1212.684
##
## Correlation Structure: General
##  Formula: ~1 | id
##  Parameter estimate(s):
##  Correlation:
##   1     2     3
## 2 0.571
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## 4 0.577 0.582 0.581
## Variance function:
##  Structure: Different standard deviations per stratum
##  Formula: ~1 | time
##  Parameter estimates:
##       y0       y1       y4       y6
## 1.000000 1.325866 1.370446 1.524806
##
## Coefficients:
##                                                          Value Std.Error
## (Intercept)                                             23.646 1.0830657
## relevel(as.factor(time), 4)y0                            2.626 0.8885231
## relevel(as.factor(time), 4)y1                            1.014 0.9343055
## relevel(as.factor(time), 4)y4                            0.424 0.9464420
## relevel(as.factor(trt), 2)A                             -2.884 1.5316862
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A  3.152 1.2565614
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A -8.254 1.3213074
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A -5.672 1.3384711
##                                                           t-value p-value
## (Intercept)                                             21.832470  0.0000
## relevel(as.factor(time), 4)y0                            2.955466  0.0033
## relevel(as.factor(time), 4)y1                            1.085298  0.2785
## relevel(as.factor(time), 4)y4                            0.447994  0.6544
## relevel(as.factor(trt), 2)A                             -1.882892  0.0605
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A  2.508433  0.0125
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A -6.246843  0.0000
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A -4.237671  0.0000
##
```

```
##  Correlation:
##                                                                (Intr) r(.(),4)0
## relevel(as.factor(time), 4)y0                                  -0.758
## relevel(as.factor(time), 4)y1                                  -0.573  0.624
## relevel(as.factor(time), 4)y4                                  -0.547  0.607
## relevel(as.factor(trt), 2)A                                    -0.707  0.536
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A  0.536 -0.707
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A  0.405 -0.441
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A  0.387 -0.429
##                                                                r(.(),4)1 r(.(),4)4
## relevel(as.factor(time), 4)y0
## relevel(as.factor(time), 4)y1
## relevel(as.factor(time), 4)y4                                   0.767
## relevel(as.factor(trt), 2)A                                     0.405     0.387
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A -0.441    -0.429
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A -0.707    -0.542
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A -0.542    -0.707
##                                                                r(.()2 r(.(),4)02
## relevel(as.factor(time), 4)y0
## relevel(as.factor(time), 4)y1
## relevel(as.factor(time), 4)y4
## relevel(as.factor(trt), 2)A
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A -0.758
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A -0.573  0.624
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A -0.547  0.607
##                                                                r(.(),4)12
## relevel(as.factor(time), 4)y0
## relevel(as.factor(time), 4)y1
## relevel(as.factor(time), 4)y4
## relevel(as.factor(trt), 2)A
## relevel(as.factor(time), 4)y0:relevel(as.factor(trt), 2)A
## relevel(as.factor(time), 4)y1:relevel(as.factor(trt), 2)A
## relevel(as.factor(time), 4)y4:relevel(as.factor(trt), 2)A  0.767
##
## Standardized residuals:
##        Min         Q1        Med         Q3        Max
## -2.1977297 -0.6919512 -0.1530936  0.5347909  5.6899331
##
## Residual standard error: 4.972081
## Degrees of freedom: 400 total; 392 residual
```

```r
# This shows the coefficients for each time point and treatment group.
# The interaction coefficients show how the treatment effect differs at each time point.

# 2. Get predicted values for each group at each time point

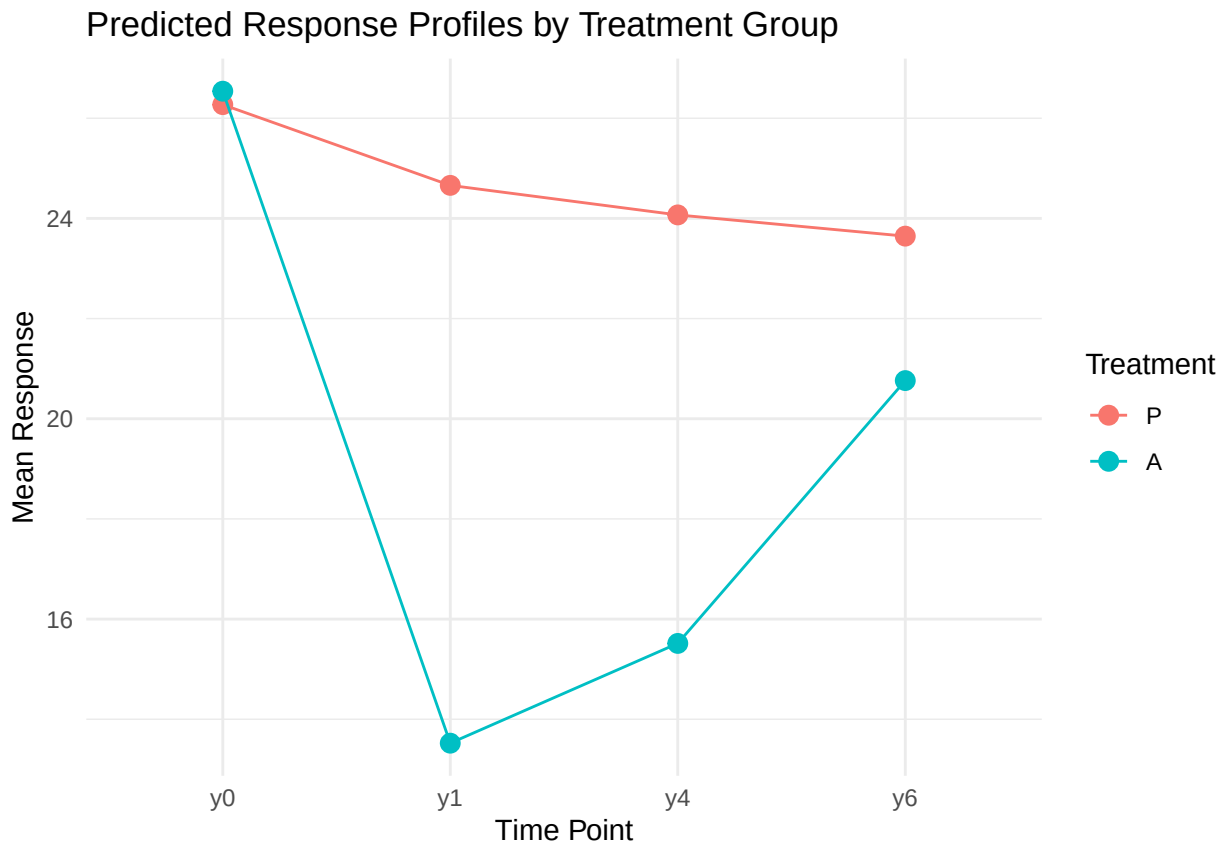
# Create a data frame with all combinations of time and treatment
newdata <- expand.grid(
    time = c("y0", "y1", "y4", "y6"),
    trt = c("P", "A")
)

# Get predictions
predictions <- predict(model1, newdata = newdata)
```

```r
# Combine predictions with the newdata
results <- cbind(newdata, predicted_y = predictions)
print(results)
```

```
##   time trt predicted_y
## 1   y0   P      26.272
## 2   y1   P      24.660
## 3   y4   P      24.070
## 4   y6   P      23.646
## 5   y0   A      26.540
## 6   y1   A      13.522
## 7   y4   A      15.514
## 8   y6   A      20.762
```

```r
# 3. Visualize the trajectories
ggplot(results, aes(x = time, y = predicted_y, color = trt, group = trt)) +
    geom_point(size = 3) +
    geom_line() +
    labs(title = "Predicted Response Profiles by Treatment Group",
         x = "Time Point",
         y = "Mean Response",
         color = "Treatment") +
    theme_minimal()
```



The output of the SAS graph is very similar to the one here in R - the two groups start out nearly identical at y0, but the active group sees a massive dip, then a strong increase immediately after the treatment is applied. The placebo group, on the other hand, has a slow decrease in the average response rate of the group as time goes on.

The groups are significantly different, which means the treatment does have an effect.